

TUSHAR JAIN

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EDUCATION

BMS College of Engineering (BMSCE), Bengaluru
B.E. Computer Science & Business Systems (CSBS)

2024 – 2028

PROJECTS

Jaipur Ride — Public Transit App

React.js, JavaScript, Vercel

- Built and shipped an offline-first metro route planner to Google Play Store; reached 1,000+ downloads and 250+ monthly active users
- Implemented client-side BFS pathfinding over a compressed JSON graph of all Jaipur Metro stations, delivering sub-10ms route calculations without network calls
- Optimized bundle to under 200KB with static data compilation, enabling reliable use on low-end devices in underground stations with no connectivity
- Managed full Play Store release cycle including privacy policies, store listing optimization, and production deployment

NotesCSBS — Academic Resource Platform

React, Vite, Supabase, Vercel

- Co-built a department-wide notes repository serving BMSCE CSBS students across all 8 semesters with lecture notes, PYQs, and study materials
- Integrated Supabase for document storage, authentication, and real-time data sync handling concurrent access during peak exam periods
- Deployed on Vercel with Google Site Verification and structured SEO; indexed by Google with organic search traffic from "BMSCE CSBS notes"

PulsePredict (Vital Health Tech) — Co-Founder & Tech Lead

ESP32, C/C++, MPU6050, MAX30102, BLE

- Designed an embedded wearable health monitor targeting elderly care at sub-₹6,000 cost; pitched at JIC Accelerator 7.0
- Wrote C++ firmware for ESP32 implementing threshold-based fall detection using accelerometer vector-sum analysis with moving-average noise filtering
- Integrated MAX30102 pulse oximeter over I2C for continuous SpO2 and heart-rate monitoring; transmitted alerts via BLE to a companion Android app
- Architected real-time patient telemetry sync to Supabase via REST APIs for remote family/doctor monitoring

BookMySlot — Multi-Tenant Reservation SaaS

HTML/CSS/JS, Supabase, Google Sheets API

- Built a 3-dashboard reservation system (Customer Booking → Staff View → Owner Dashboard) with Supabase Realtime for instant cross-device sync
- Implemented PostgreSQL Row-Level Security for tenant isolation and pessimistic locking to prevent double-booking race conditions
- Integrated Google Sheets API as a lightweight fallback for non-technical restaurant owners accustomed to spreadsheet workflows

Split-Payment System — FinTech Prototype

JavaScript, Node.js, Razorpay API, REST

- Engineered a multi-source checkout splitting a single bill across UPI and Card with Razorpay test-mode integration
- Implemented atomic transaction logic with rollback safety — if either payment stream fails, the system auto-refunds the captured charge
- Designed retry-safe idempotency checks preventing duplicate charges on page reload or network retry

Air Guitar — Gesture-Controlled Synthesizer

Python, Arduino, MPU6050, NumPy, Serial

- Built a real-time virtual guitar using MPU6050 accelerometer on a glove, streaming sensor data at 115200 baud to a Python audio engine
- Implemented Karplus-Strong string synthesis algorithm producing acoustic guitar tones with sub-10ms latency

RESEARCH

A Conceptual Framework for a Human-Supervised Dual-UUV System for Autonomous Maritime Surveillance

2025 · Published on ResearchGate

- Designed a dual-agent UUV architecture using Extended Kalman Filter sensor fusion for underwater navigation and cooperative path planning
- Established a human-in-the-loop governance model requiring cryptographic approval before any autonomous intercept action

HACKATHONS

Safe Route — Top 10 Finalist out of 500+ teams

- Built a safety-optimized route recommendation system during a competitive hackathon

SKILLS

Languages: JavaScript, TypeScript, C/C++, Python, SQL, HTML, CSS

Frameworks: React, Next.js, Vite, Node.js, Express, Tailwind CSS

Backend & Databases: Supabase (PostgreSQL), Firebase, MySQL, REST APIs, Razorpay API, Google Sheets API

Embedded & IoT: ESP32, Arduino, MPU6050, MAX30102, BLE, I2C/SPI/UART

Developer Tools: Git, GitHub, Vercel, Netlify, VS Code, Figma, Postman, Arduino IDE

CERTIFICATIONS

Investment Risk Management — Coursera (Verify: MWIVK1DVGR6J)

Business Analysis & Process Management — Coursera (Verify: 76W901BZ2HPB)